

* Consumer Behaviour $\rightarrow$ Preference and choice.

$\rightarrow$ Demand Theory

$\rightarrow$ Production Theory

$\rightarrow$ General and Welfare economics

* Externality and Public goods.

* Asymmetric Information

Chap-1

Preferences and Utility

$\rightarrow$ Distinguish between preference and choice

$\rightarrow$ कौनसा Rational consumer कौनसे छेले छेले?

Assumption:

$\rightarrow$ convexity $\rightarrow$ quasi concavity $\rightarrow$ continuous and differentiable

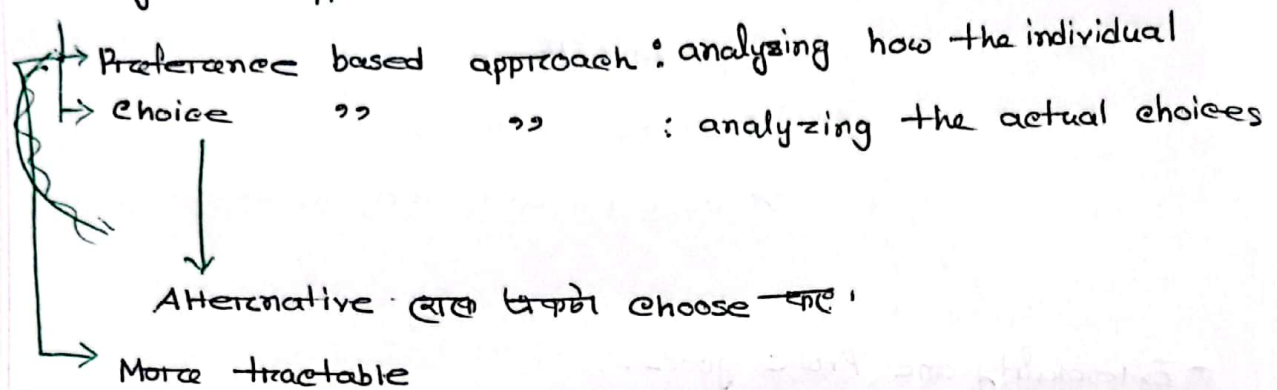
$\rightarrow$ Social and Reference $\rightarrow$ Hyperbolic and quasi hyperbolic $\rightarrow$ Revealed

preference $\begin{cases} \rightarrow \text{Strong axiom} \\ \rightarrow \text{Weak} \end{cases}$

$\rightarrow X \subseteq \mathbb{R}_+^N$ $\rightarrow$ Positive
 $\downarrow$
 Possible alternative bundles of commodities

$X = \{x, y, z, \dots\}$
 $\downarrow \downarrow \downarrow$
 commodity

2 ways to approach the decision making process.



सावधानी (Better) $\rightarrow$ Preference $\cup$ अच्छा Bundle - सावधानी affordability व सार्वजनिक choice afford करते हैं।

Rational preferences $\Rightarrow$ (implies) consistent choice behaviour.

Rational preferences $\Leftarrow$ (implied by) consistent choice behaviour.

$\rightarrow$ Preference is an attitude of the decision maker towards the possible bundle of X . (Compare करना ability - सावधानी से)

$$x, y \in X$$

$$x \succ y \text{ (पसंद } x \text{ to } y)$$

$$y \succ x \text{ (पसंद } y \text{ to } x)$$

$$x \sim y \text{ (} x \text{ indifferent to } y)$$

$\rightarrow$ Compare करना ability ना सावधानी rational ना $\rightarrow$ rational व preference complete $>$ completeness assumption.

x व y जो $\sim$ नहीं यदि Binary relationship सावधानी may not satisfy completeness assumption. Not all Binary relationship satisfy completeness

→ Commodities ~~are~~ are to be comparable → Binary relationship.

• प्रत्येक प्रकार complete रहे।

→ Strong preference - साफ रहे।

चर: 1.1

Chap-1

→ Preference-Based Approach:

Reflexivity: $x \in X \Rightarrow$ 1) $x \sim x$ 2) $x \succeq x$ 3) $x \not\succeq x$

$\downarrow$ $\downarrow$ $\downarrow$
 indifferent weakly preferred not strongly preferred

* Who is the rational consumer?

→ Completeness and transitivity fulfill $\Rightarrow$

$\downarrow$

$x \succeq y, y \succeq z \Rightarrow x \succeq z$

* Preference relationship: $x \succeq y$ if and only if $\sum_{i=1}^N x_i \geq \sum_{i=1}^N y_i$ (1.1)

→ Completeness assumption
 → Transitivity → key assumption

(1.1) continues

* Example violating Transitivity:

Transitivity assumptions should not cycle.

$\downarrow$

Why this cycle happen? → sources of intransitivity:

- ① → Indistinguishable alternative
- ② → Framing effects
- ③ → Aggregation of criteria
- ④ → Change in preference

① Volcanian difference → Ordered pairs of x and y difference.
 $\downarrow$
 1 x and y are indistinguishable.

② Cycle x and y (Condorcet paradox) → Every person in the group has different preference but group preferences are not necessarily transitive

Utility Function

→ IE - Ordinal measure → Law of Diminishing marginal utility - cardinal

$$U: X \rightarrow \mathbb{R}, x, y \in X; x \succeq y \Leftrightarrow U(x) \geq U(y)$$

↓
necessary and sufficient

Two points: ① only the ranking of alternative matters, cardinality is not cared.

② strictly increasing function $f(\cdot)$ on $U(x)$ i.e., → new function has the same preference.

→ Desirability:

→ Desirability:

→ Monotonicity: More is preferred to less (Higher IC)

→ Strong monotonicity.

→ Non-satiation: Non satisfaction

→ Local non-satiation

bliss point → satiation point

$x \in \mathbb{R}$ i.e. $x = (x_1, x_2, \dots, x_n)$

x_k ;

Monotonicity: a) $x_k \geq y_k \Rightarrow x \succeq y$ (weakly preferred) b) $x_k > y_k \Rightarrow x \succ y$ (strongly preferred)

Strong Monotonicity: $x_k \geq y_k \Rightarrow x \succ y$ * Under monotonicity we can increase the all but one good.

→ Monotonicity → अब बाजार हम strong प्र. करे। strong प्र. एकटे बा-रह।

< हमने जो Utility function बाजार >

Example 1.5 Extreme → perfect complementary → Monotonicity fulfill.

किंतु strong monotonicity नहीं है।

Example 1.6 Perfect substitute → monotonicity + strong monotonicity

रखे hold करे।

Strong monotonicity implies monotonicity but not vice versa

Non satiation: $x \in X, y \in X ; y \succ x \rightarrow x$ पर काइकाहि आलेकटे good आले बाटे x पर लेले जाय।

Local non-satiation: $\Rightarrow x \succ y$ पर लेले काइ दूरे?

A preference relation satisfies LNs if $x \in X, \epsilon > 0$; another bundle

$y \in X$, which is less than ϵ away from x , $\|y-x\| < \epsilon$ and $y \succ x$

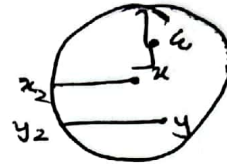
→ यदि ϵ circle consider करें तो $\epsilon \rightarrow$ radius



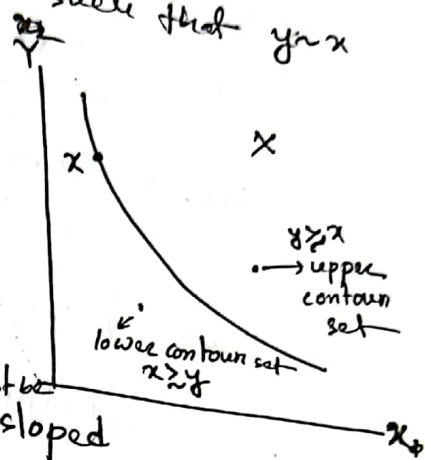
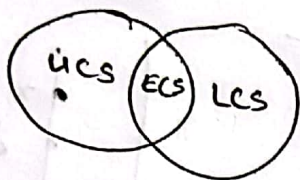
$$\|y-x\| = \sqrt{(y_1-x_1)^2 + (y_2-x_2)^2}$$

→ A preference relation satisfies

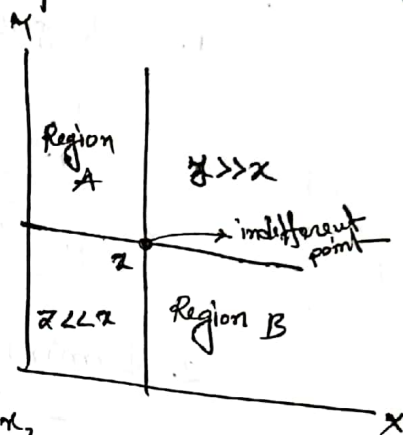
→ २२ २२ के (Economic Bad good)



Violation of LNS



- ✓ strong monotonicity $\rightarrow$ IC negatively ^{must be} sloped
- ✓ LNS implies $\rightarrow$ IC can be upward sloping.



- convexity of preference:
for all

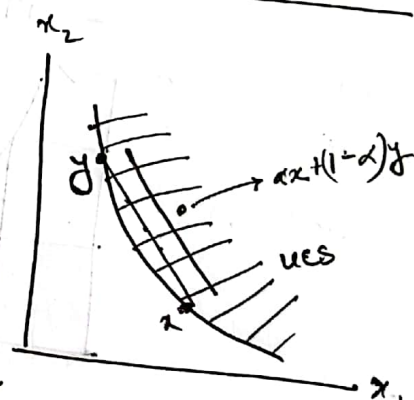
for all $x, y \in X$

$$x \sim y \Rightarrow \text{for } \alpha \in [0, 1] \quad \alpha x + (1 - \alpha)y$$

२. ७५ - १३ - माहुर धातु (रक्षा) धमीकरण
अमे $\alpha x + (1-\alpha)y$

✓ any point on the 'strong' is weakly preferred.

but x & y are also two point on the straight line.



✓ So we can call it ~~not~~ strictly convex

but if x & y are not on $\mathbb{R}$

✓ Strictly convexity needed to get one unique.
equ^m point

✓ convex but not strictly convex preference

For substitutes

— convex but not strictly convex

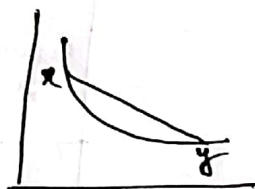
For complement

— convex but not strict convexity

Table of convexity vs strict convexity

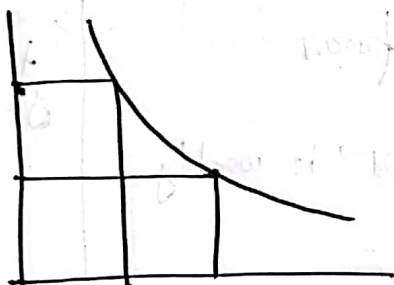
• Interpretation of convexity

1. Taste for diversification



convex combination of bundles x & y
than either x or y

2. Diminishing Marginal rate of substitution
— MRS has to be diminishing



→ slope for MRS : u

Indifference $f^u : u(x_1, x_2)$

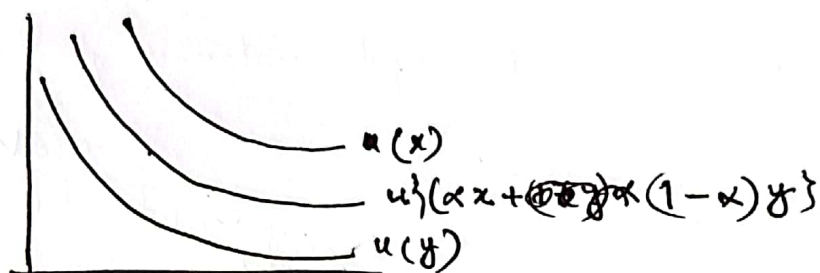
for slope → total differential : $du = \frac{\partial u}{\partial x_1} dx_1 + \frac{\partial u}{\partial x_2} dx_2 + \dots +$

$$\frac{\partial u}{\partial x_n} dx_n = 0$$

- Quasiconcavity: if every bundle lies is convex $\Rightarrow u(\cdot)$ quasiconcave.

There's x & y bundle
 $\min u(x) \text{ \& } \min u(y)$ separately $\leq$ convex combination
 of these two bundles

$$u(\alpha x + (1-\alpha)y) \geq \min \{u(x), u(y)\}$$



- ✓ strict quasiconcavity
- ✓ Concavity vs quasiconcavity
- Common type of utility f^n :
 - ✓ Cobb-Douglas utility $f^n \rightarrow$ LR utility f^n
 $x_1 \text{ \& } x_2$
 $\alpha, \beta \rightarrow$ partial elasticity.
 - ✓ Monotonic transformation of Cobb-Douglas.
 - ✓ For Perfect substitute
 - ✓ Slope same $\rightarrow$ monotonic transformation

- CES utility f^n
- Quasilinear f^n — $1/2$ linear $1/2$ non-linear
- Properties of Preference Relations:
 - Homogeneity
 - Lexicographic preference

→ Utility maximization problem:

$$\max_{x \geq 0} U(x)$$

$$\text{s.t. } p \cdot x \leq \omega \quad \rightarrow \text{budget constraint}$$

Total expenditure on consumption bundle x

→ Weierstrass Theorem:

bounded set: Consumer goes for this set of commodity follows

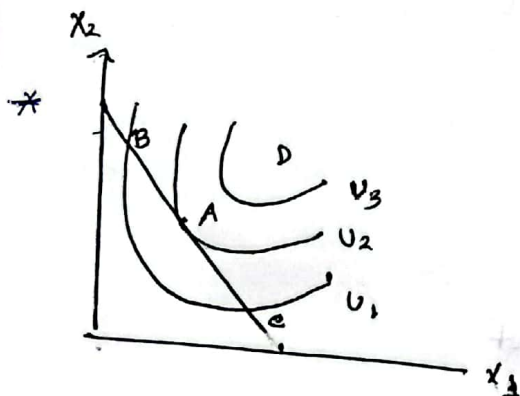
closed " : @ " concentrates on this bundle

Existence: if $p \gg 0$, $\omega > 0$, $U(\cdot)$ argument of x → continuous, then at least one solution. If the preferences are strictly convex, then the solution is unique.

argmax → argument x that solves the optimization problem $x(p, \omega)$.

* Marshallian demand γ_s ← Walrasian demand correspondence

Same but M.D to γ_s income ω to income + wealth



Utility maximization → given the consumer's wealth

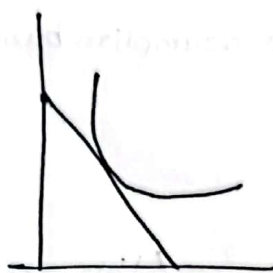
Expenditure minimization → utility constant

→ Properties of Walrasian Demand

□ Homogeneity of degree zero

□ Walras' Law

□ Convexity / Uniqueness → Convex set → upper contour set
 Strictly convex & Unique sol, वा सय अवस्थिति पाता



→ UPM Problem: Necessary Condition

$$\text{Max } U(x) \quad \text{s.t. } p \cdot x \leq w \\ x \geq 0$$

$U = x_1 x_2$	$u = \sqrt{x_1 x_2}$	$u = \frac{1}{2} \ln x_1 x_2$
$p_1 x_1 + p_2 x_2 = w$	$p_1 x_1 + p_2 x_2 = w$	$\rightarrow \text{math } \bar{x}_1, \bar{x}_2$
Solve: $\bar{x}_1 = \frac{w}{2p_1}, \bar{x}_2 = \frac{w}{2p_2}$		

Sufficient Condition

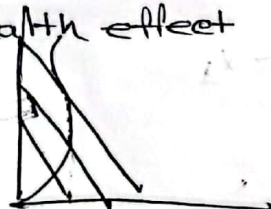
→ Violations of the sufficient condition

□ $u(\cdot)$ is monotone

□ $x \succ y$ &

→ Corner solution

→ Walrasian demand: wealth effect



→ Engel curve

→ Walrasian Demand: Price effect
~~for~~

→ Indirect utility function: $v(p, y)$ CR → value function from maximization problem

→ Properties of Indirect utility function

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असु

→ Engel curve

→ Hicksian Demand: Price effect
for

→ Indirect utility function: $v(p, y)$ CR → value function from maximization problem

→ Properties of Indirect utility function

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Expenditure Maximization Problemmax $U(x)$

s.t.

$$x \in U(p) \quad \text{A new level of } U$$

3a

Expenditure

Max $x_1, x_2, x_3, \dots$ s.t. $x_1, x_2, x_3, \dots$

All other things stable demand function

where $(x_1, x_2, x_3, \dots)$

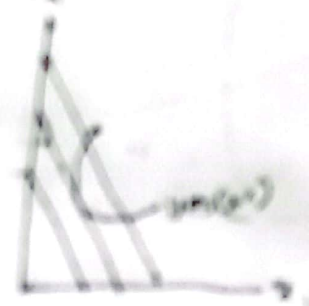
$$\text{Expenditure } U_1 = \sqrt{\frac{x_1^2}{p_1^2}}, \quad U_2 = \sqrt{\frac{x_2^2}{p_2^2}}$$

as the demand p is $p_1 > p_2 > p_3$ early income variations

Prove that

A better demand level p is better than p_1

Expenditure



→ Solve all the Expenditure maximization problems

$$\text{Max } \frac{x_1}{p_1} = \frac{x_2}{p_2} = \dots \rightarrow$$

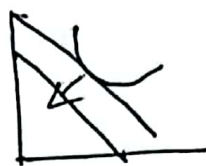
$$\frac{x_1}{p_1} = \frac{x_2}{p_2} \rightarrow \text{Budget share}$$

Consumption = 11.5

Expenditure Maximization ProblemMax $U(x)$ s.t. $x \in U(p)$ Max $U(x)$

→ Properties of Hicksian Demand

1. Homog (0)
2. No excess utility
3. Convexity
4. Uniqueness



→ The Expenditure function

$$p \cdot h(p, u) = e(p, u)$$

Properties: 1. Homog (1) 2. strictly increasing in u , 3. Non-decreasing in p_k for every good k 4. Concave in p .

→ Relationship between the Expenditure and Hicksian demand

Shepard's lemma →

Demand f^n from indirect utility function → Roy's Identity

Profit π from input demand function → Hotelling's Lemma

→ Relationship between Hicksian and Walrasian demand

↓

~~Substitution~~

Substitution Effect

↓

Income substitution

SY Effect

→ Summary Relationship

Shepard's lemma, Roy's Identity, Slutsky equation,

Quart
2022

Slutsky eqⁿ → income & substitution effect

Quasilinear - 1 part linear & 1 part non-linear

21.04.24

21 May - Tutorial (9AM - 12PM)

→ Define prodⁿ Set & graphically explain its properties with notations (Chp-1).

Chapter - 3 : Demand Theory & Application

- measuring consumer's welfare → consumer surplus.
- Welfare Evaluation: Compensation & Equivalent variation
- Quasilinear preferences
- Slutsky Eqⁿ
- Gross & net substitutability

□ How to measure welfare effects of

- Price decrease/increase
- introduction of tax/subsidy

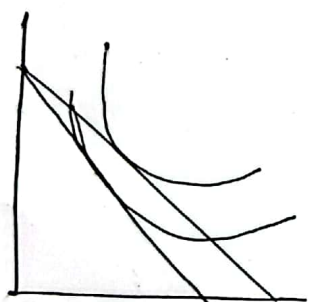
price ↑ / price ↓

• Why not use difference in utility?

- within a subject criticism:

- Betⁿ a u u

• Price decrease:



• Money Metric evaluation

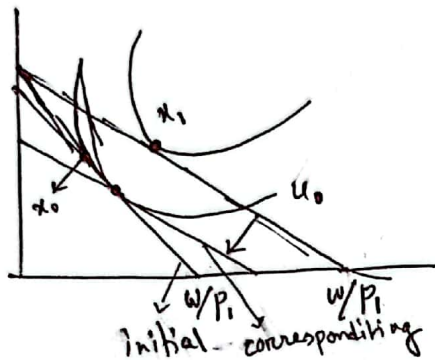
- Compensating Variation:
- Equivalent variation:

Two approach

- Using Expenditure f^m :
- Using the Hicksian demand f^h

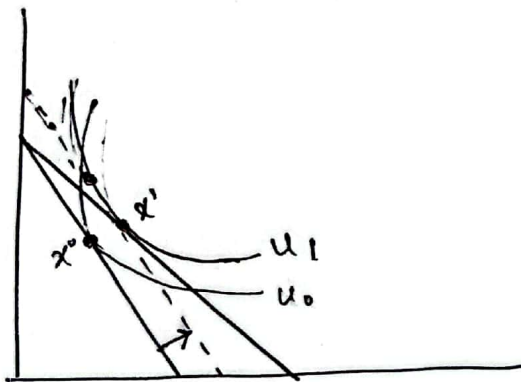
CV using expenditure f^h

$$CV(P^0, P^1, w) = e(P^1, u^0) - e(P^0, u^0)$$

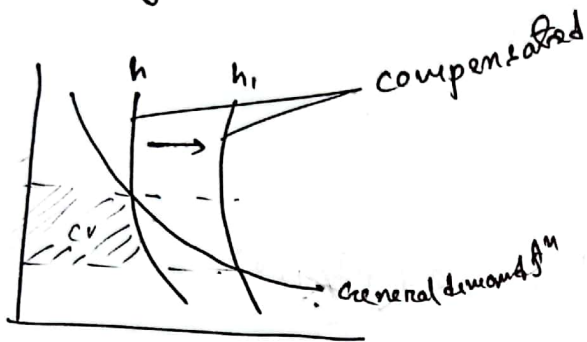


EV using expenditure f^m

$$EV(P^0, P^1, w) = e(P^0, u_1) - e(P^0, u^0)$$



CV using Hicksian



Why not using Walrasian demand?

- Area Variation

Gross & Net substitutes